Testing Document

Study Planner / Assignment Tracker

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1. Test Plan

1.1 Testing Objectives

- Verify all 8 functional requirements are correctly implemented.
- Validate input boundary conditions and invalid input handling.
- Confirm API endpoints return correct HTTP status codes and response shapes.
- Ensure data persistence and integrity across operations.

1.2 Testing Strategy

The test suite uses a **multi-level approach**:

Level	Scope	Tool	Location
Unit	Individual classes and functions	pytest	`tests/test_models.py`, `tests/test_repository.py`, `tests/test_service.py`
Integration	Flask API endpoints (full request/response cycle)	pytest + Flask test client	`tests/test_api.py`
Manual (exploratory)	Full UI interaction in browser	Manual	localhost:5173

1.3 Types of Testing Applied

- **Functional testing** — verifies that each feature works as specified (happy path)
- **Boundary value analysis** — tests values at the edges of valid input ranges (e.g., blank strings, impossible dates, exactly-48-hour window)
- **Equivalence class testing** — tests one representative from each valid and invalid input partition (e.g., a valid priority vs. an invalid priority)
- **Error path testing** — verifies correct HTTP error codes and messages for invalid or missing resources

1.4 Test Environment

Component	Version
Python	3.13.7
pytest	8.2.2
Flask	3.0.3

Component	Version
flask-cors	4.0.1
Node.js	18+
OS	Windows 11

1.5 Test Data Management

- All backend tests use `pytest`'s `tmp_path` fixture to redirect file I/O to a temporary directory.
- The real `backend/assignments.json` is **never touched** by any automated test.
- Each test function starts with an empty data store (fixture is function-scoped).

1.6 Running the Tests

```
# From the project root
python -m pytest tests/ -v
```

Expected result: **27 passed**.

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2. Test Cases

2.1 Model Tests (T01–T04)

ID	Test Name	Input	Expected Output	Test Type
T01	`test_from_dict_valid`	Complete valid dict with all 7 fields	Assignment object with matching field values	Functional
T02	`test_to_dict_roundtrip`	Assignment → to_dict() → from_dict()	Resulting object is identical to original	Functional
T03	`test_default_completed_false`	Dict without `completed` key	Assignment.completed == False	Boundary (default value)
T04	`test_to_dict_contains_all_fields`	Any valid Assignment	to_dict() result has exactly {id, title, course, due_date, priority, completed, created_at}	Contract

2.2 Repository Tests (T05–T08)

ID	Test Name	Input	Expected Output	Test Type
T05	`test_save_and_get_all`	Save one Assignment, then get_all()	List with exactly that one Assignment	Functional

ID	Test Name	Input	Expected Output	Test Type
T06	`test_update_existing`	Save same ID twice with different title	get_all() returns 1 record with new title	Functional (idempotency)
T07	`test_delete_existing`	Save then delete by ID	get_all() returns empty list; delete returns True	Functional
T08	`test_delete_nonexistent_returns_false`	Delete an ID that was never saved	Returns False; original record untouched	Boundary (not found)

2.3 Service Tests (T09–T19)

ID	Test Name	Input	Expected Output	Test Type
T09	`test_validate_missing_title`	Data dict with no `title` key	Raises ValueError mentioning "title"	Equivalence (invalid input)
T10	`test_validate_blank_title`	title = `"  "` (whitespace only)	Raises ValueError mentioning "title"	Boundary (empty after strip)
T11	`test_validate_invalid_priority`	priority = `"urgent"`	Raises ValueError mentioning "priority"	Equivalence (invalid enum value)
T12	`test_validate_invalid_date_format`	due_date = `"03/21/2026"` (MM/DD/YYYY)	Raises ValueError mentioning "due_date"	Equivalence (wrong format)
T13	`test_validate_impossible_date`	due_date = `"2026-02-30"`	Raises ValueError mentioning "due_date"	Boundary (non-existent date)
T14	`test_create_generates_uuid`	Two separate valid create calls	Both have non-null IDs; IDs are different	Functional
T15	`test_get_assignments_filter_by_course`	3 assignments (2 CS101, 1 MATH201); filter="CS101"	Returns exactly 2 assignments, all course="CS101"	Functional (FR-07)
T16	`test_get_assignments_sort_by_due_date`	3 assignments with dates: 2026-05-01, 2026-03-25, 2026-04-10; sort="due_date"	Returned in order: 03-25, 04-10, 05-01	Functional (FR-06)

ID	Test Name	Input	Expected Output	Test Type
T17	`test_is_upcoming_soon_within_48h`	Incomplete assignment due in 24 hours	Returns True	Boundary (within window)
T18	`test_is_upcoming_soon_beyond_48h`	Incomplete assignment due in 72 hours	Returns False	Boundary (outside window)
T19	`test_is_upcoming_soon_completed_ignored`	Completed assignment due in 1 hour	Returns False	Business rule (FR-08)

2.4 API Tests (T20–T27)

ID	Test Name	HTTP Request	Expected Status	Expected Body	Test Type
T20	`test_get_assignments_empty`	GET /api/assignments (empty store)	200	`{"assignments": []}`	Functional
T21	`test_post_assignment_valid`	POST with valid payload	201	Contains `id`, `title`, `completed: false`	Functional (FR-01)
T22	`test_post_assignment_missing_field`	POST with no `title`	400	Contains `"error"` key	Error path
T23	`test_put_assignment_valid`	PUT existing ID with changed title	200	Updated title in response	Functional (FR-03)
T24	`test_put_assignment_not_found`	PUT unknown ID	404	Contains `"error"` key	Error path
T25	`test_delete_assignment_valid`	DELETE existing ID	200	Assignment gone on subsequent GET	Functional (FR-05)
T26	`test_delete_assignment_not_found`	DELETE unknown ID	404	Contains `"error"` key	Error path
T27	`test_patch_toggle_complete`	PATCH /complete on incomplete → then again	200 twice	completed flips true then false	Functional (FR-04)

3. Test Results

All 27 tests pass:

```
tests/test_api.py::test_get_assignments_empty           PASSED
tests/test_api.py::test_post_assignment_valid           PASSED
tests/test_api.py::test_post_assignment_missing_field   PASSED
tests/test_api.py::test_put_assignment_valid            PASSED
tests/test_api.py::test_put_assignment_not_found        PASSED
tests/test_api.py::test_delete_assignment_valid         PASSED
tests/test_api.py::test_delete_assignment_not_found     PASSED
tests/test_api.py::test_patch_toggle_complete           PASSED
tests/test_models.py::test_from_dict_valid              PASSED
tests/test_models.py::test_to_dict_roundtrip            PASSED
tests/test_models.py::test_default_completed_false      PASSED
tests/test_models.py::test_to_dict_contains_all_fields  PASSED
tests/test_repository.py::test_save_and_get_all         PASSED
tests/test_repository.py::test_update_existing          PASSED
tests/test_repository.py::test_delete_existing          PASSED
tests/test_repository.py::test_delete_nonexistent_returns_false PASSED
tests/test_service.py::test_validate_missing_title      PASSED
tests/test_service.py::test_validate_blank_title        PASSED
tests/test_service.py::test_validate_invalid_priority   PASSED
tests/test_service.py::test_validate_invalid_date_format PASSED
tests/test_service.py::test_validate_impossible_date    PASSED
tests/test_service.py::test_create_generates_uuid       PASSED
tests/test_service.py::test_get_assignments_filter_by_course PASSED
tests/test_service.py::test_get_assignments_sort_by_due_date PASSED
tests/test_service.py::test_is_upcoming_soon_within_48h PASSED
tests/test_service.py::test_is_upcoming_soon_beyond_48h PASSED
tests/test_service.py::test_is_upcoming_soon_completed_ignored PASSED

27 passed in 0.83s
```

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4. Requirements Traceability Matrix

The following matrix maps each functional requirement to the test cases that verify it.

Requirement	Description	Test IDs	# Tests
FR-01	Add assignment	T09, T10, T11, T12, T13, T14, T21, T22	8
FR-02	View assignment list	T05, T15, T16, T20	4
FR-03	Edit assignment	T06, T23, T24	3
FR-04	Mark assignment complete	T27	1
FR-05	Delete assignment	T07, T08, T25, T26	4
FR-06	Sort by due date	T16	1
FR-07	Filter by course	T15	1
FR-08	Upcoming soon warning	T17, T18, T19	3

Total coverage: All 8 functional requirements have at least one passing test. All 27 tests are mapped to at least one requirement.

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5. Boundary Value Analysis

5.1 Title Field (string, required)

Partition	Test Value	Class	Expected
Valid	`"Chapter 5 Review"`	Valid non-empty string	Accepted
Invalid — missing	(key absent)	Missing required field	ValueError (T09)
Invalid — blank	`" "`	Whitespace-only string	ValueError (T10)

5.2 Priority Field (enum)

Partition	Test Value	Class	Expected
Valid	`"low"`, `"medium"`, `"high"`	Valid enum member	Accepted
Invalid	`"urgent"`, `"1"`, `""`	Outside enum	ValueError (T11)

5.3 Due Date Field (YYYY-MM-DD)

Partition	Test Value	Class	Expected
Valid	`"2026-04-01"`	Correct format, real date	Accepted
Invalid format	`"03/21/2026"`	Wrong separator	ValueError (T12)
Invalid date	`"2026-02-30"`	Correct format, impossible date	ValueError (T13)

5.4 Upcoming Soon Window (48-hour boundary)

Partition	Due offset	Expected
Well inside	due in 24h, incomplete	True (T17)
Well outside	due in 72h, incomplete	False (T18)
Completed override	due in 1h, completed	False (T19)

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6. Equivalence Class Summary

Input	Valid Classes	Invalid Classes
Title	Non-empty, non-blank string (any length 1–200)	Missing, blank/whitespace-only, absent key
Course	Non-empty, non-blank string (any length 1–100)	Missing, blank/whitespace-only, absent key
Due Date	Valid ISO 8601 date (YYYY-MM-DD)	Wrong format (MM/DD/YYYY, YYYY/MM/DD), impossible date
Priority	`"low"` OR `"medium"` OR `"high"`	Any string not in that set, missing key
Assignment ID (for PUT/DELETE/PATCH)	Existing UUID in data store	UUID not in store, arbitrary string